

Pratham Agrawal

+91-8269067007 — prathamagrwal1205@gmail.com — linkedin.com/in/pratham-manish-agrawal
github.com/prathamagrwal — leetcode.com/u/prathamagrwal — kaggle.com/fearsomejockey

Experience

Affinsys AI — AI B2B Startup

Software Engineer

Bangalore, India

Jan 2024 - Present

- **DBOps Service:** Designed and shipped an **LLM-driven ETL pipeline orchestrator** that converts natural language queries into executable workflows using Claude (Anthropic API) with structured prompt chaining — **reducing manual pipeline authoring from hours to under 2 minutes** for data ops teams.
- **Datalens Service:** Architected a self-serve **analytics platform enabling NLQ-to-SQL generation via LLMs**, with SQL sanitisation, metadata-driven query construction, and a chart catalog engine supporting 15+ chart types. **Reduced dashboard creation time by 90%**, enabling non-technical stakeholders to independently build and iterate on reports.
- **Analytics Service:** Led end-to-end development of a multi-tenant analytics **service powering 20+ dashboards and 200+ charts for internal and client-facing teams**. Introduced materialized SQL views to eliminate redundant query execution, reducing database load by 40% and **cutting average API response time from 800ms to 480ms**.
- **Analytics Scheduler:** Designed a scheduling engine to **automate recurring delivery of 100+ analytics reports via email, reducing manual reporting effort by an estimated 15 engineer-hours** per week.
- **Archival Service:** Implemented **comprehensive archiving solution** that guarantees data consistency, referential integrity, and reliability across a distributed system of **10+ services**, encompassing complex foreign key relationships, failover strategy, data purge algorithm, and seamless data and files (from MinIO to AWS S3) migration.
- **Eventlogger Service :** Built a high-throughput **event-driven data warehouse ingesting events from 10+ services via AMQP/NATS into 40+ structured tables** using async DB writes — achieving sub-second ingestion latency and enabling analytics queries that previously required multi-service joins to run against a single source of truth.

CME Group — World's largest derivatives exchange

SDE Intern

Bangalore, India

May 2023 - Nov 2023

- Re-engineered report and scheduled email generation pipelines for a high-frequency trading data platform, **achieving a 30% throughput improvement by refactoring legacy batch processing logic**.
- Built internal data extraction and transformation tooling that reduced pipeline latency by 40%, enabling faster downstream reporting for trading operations teams.

Projects

Mirror-DB | *Kubernetes, PostgreSQL, PgBouncer, Helm Chart*

May 2025

- **Problem:** Production PostgreSQL clusters lose availability during primary failover, causing downtime.
- Designed a HA PostgreSQL cluster on Kubernetes **achieving sub-60s automatic failover, 70–90% connection overhead reduction** via PgBouncer pooling, and read/write splitting for zero-downtime operations.
- Implemented **dynamic service discovery with automated pod labeling**, multi-tier synchronous/asynchronous replication across 4-node cluster, and comprehensive monitoring stack

LogHive | *Python, RabbitMQ, Redis, PostgresDB, Docker*

Dec 2024

- **Problem:** High-volume log systems overwhelm storage and retrieval without back-pressure management.
- Built a log archival pipeline with compression and **indexed retrieval, benchmarked at high-throughput ingestion with 99.99% write durability** on PostgresSQL.
- Implemented adaptive RabbitMQ consumer scaling and producer rate-limiting via Redis to prevent queue saturation under load spikes.

Technical Skills

Languages: Python, C++, Java, SQL

Databases: PostgreSQL, MongoDB, Oracle — schema design, query optimisation, materialized views, data warehousing

Data Engineering: Apache Airflow (DAG authoring), Apache Spark (batch processing), ETL pipeline design

Messaging & Streaming: RabbitMQ, Apache Kafka, NATS (AMQP/JetStream)

Infrastructure & Cloud: Docker, Kubernetes (Helm, pod lifecycle), AWS (S3, EC2, IAM), GitHub Actions

Observability: Prometheus, Grafana

LLM / AI: Anthropic Claude/ OpenAI API, prompt chaining, structured outputs, self-correction loops

Education

Vellore Institute of Technology, Vellore

Sep, 2020 – July 2024

Bachelors in Computer Science with specialization in Data Science

CGPA - 9.25

Certification

Google Cloud Computing Foundations Program